

Supplementary Sensitivity Analysis I

To address potential heteroscedasticity of the differences between predicted and reference maps, supplementary log-ratio Bland–Altman (multiplicative LoA) and percentage Bland–Altman (relative LoA) formulations were computed in addition to the standard absolute-kPa formulation. Distribution-free bootstrap (5000 resamples) 95% confidence intervals for CCC are reported alongside the Fisher z-transformation intervals as a complementary inferential check. All sensitivity and robustness analyses are presented in Supplementary Tables S2–S4.

All analyses use the final test cohort of 13 patients / 26 slices. Slice-level statistics use $n = 26$ slices; patient-level statistics use the mean of the two test slices per patient ($n = 13$). Bland-2007 LoA apply the within-subject variance correction (Bland & Altman, 2007) to the 26 slice-level differences. Confidence intervals for Bland–Altman LoA endpoints follow Bland & Altman (1986); Lin's CCC 95% confidence intervals via Fisher z-transformation are reported as the primary CIs; bootstrap CIs (5000 resamples) are reported in Table S3 as a complementary distribution-free estimate.

The corrected LoA (reported in this file) differed from the uncorrected slice-level LoA (reported in the manuscript) by less than 0.02 kPa across all models, indicating that within-subject correlation had a negligible effect on the slice-level agreement estimates.

TABLE S2. SLICE-LEVEL AND PATIENT-LEVEL AGREEMENT ANALYSES FOR STORAGE MODULUS (GD) ON THE TEST SET (13 PATIENTS / 26 SLICES).

Model	Slice bias (kPa)	Slice LoA (kPa)	Patient bias (kPa)	Patient LoA (kPa)	Bland-2007 LoA (kPa)	Slice CCC [95% CI]	Patient CCC [95% CI]
Random init.	+0.127	[-1.51, +1.77]	+0.127	[-1.34, +1.60]	[-1.53, +1.79]	0.896 [0.797, 0.949]	0.917 [0.793, 0.968]
AE (Full Finetuning)	+0.136	[-1.20, +1.47]	+0.136	[-1.09, +1.37]	[-1.21, +1.49]	0.941 [0.874, 0.973]	0.951 [0.849, 0.985]
AE (Warmup-PerBlockLR)	+0.114	[-1.08, +1.31]	+0.114	[-1.08, +1.31]	[-1.10, +1.33]	0.950 [0.894, 0.977]	0.952 [0.855, 0.985]
AE (Warmup-LRReset)	+0.119	[-1.07, +1.31]	+0.119	[-1.05, +1.29]	[-1.09, +1.33]	0.953 [0.899, 0.979]	0.956 [0.863, 0.986]
AE (Fixed)	+0.109	[-1.01, +1.22]	+0.109	[-1.00, +1.22]	[-1.03, +1.24]	0.958 [0.910, 0.981]	0.960 [0.877, 0.988]
Rot (Full Finetuning)	+0.078	[-0.95, +1.11]	+0.078	[-0.89, +1.05]	[-0.97, +1.12]	0.961 [0.923, 0.980]	0.966 [0.913, 0.987]
Rot (Warmup-PerBlockLR)	+0.008	[-1.21, +1.22]	+0.008	[-1.07, +1.09]	[-1.22, +1.24]	0.945 [0.894, 0.971]	0.957 [0.901, 0.982]
Rot (Warmup-LRReset)	+0.046	[-1.15, +1.24]	+0.046	[-1.00, +1.09]	[-1.16, +1.26]	0.947 [0.896, 0.973]	0.960 [0.904, 0.984]
Rot (Fixed)	+0.019	[-1.28, +1.32]	+0.019	[-1.10, +1.14]	[-1.29, +1.33]	0.936 [0.880, 0.967]	0.953 [0.896, 0.979]

For each model, the table reports slice-level Bland–Altman ($n=26$), patient-level Bland–Altman ($n=13$, mean of two test slices per patient), the Bland & Altman 2007 within-subject-corrected LoA computed on the 26 slice-level differences, and Lin's concordance correlation coefficient (CCC) at both slice and patient levels with 95% CIs via Fisher z-transformation. The slice-level data complement the patient-level results presented in Table V of the main paper.

TABLE S3. SLICE-LEVEL AND PATIENT-LEVEL AGREEMENT ANALYSES FOR LOSS MODULUS (GL). COLUMNS ARE AS IN TABLE S2.

Model	Slice bias (kPa)	Slice LoA (kPa)	Patient bias (kPa)	Patient LoA (kPa)	Bland-2007 LoA (kPa)	Slice CCC [95% CI]	Patient CCC [95% CI]
Random init.	+0.102	[-1.36, +1.57]	+0.102	[-1.16, +1.37]	[-1.38, +1.58]	0.835 [0.713, 0.908]	0.877 [0.760, 0.939]
AE (Full Finetuning)	+0.118	[-0.81, +1.04]	+0.118	[-0.72, +0.95]	[-0.82, +1.05]	0.942 [0.884, 0.971]	0.953 [0.874, 0.983]
AE (Warmup-PerBlockLR)	+0.089	[-0.79, +0.97]	+0.089	[-0.77, +0.95]	[-0.81, +0.99]	0.945 [0.901, 0.970]	0.949 [0.882, 0.979]
AE (Warmup-LRReset)	+0.108	[-0.67, +0.88]	+0.108	[-0.63, +0.85]	[-0.68, +0.90]	0.958 [0.920, 0.978]	0.962 [0.906, 0.985]
AE (Fixed)	+0.088	[-0.70, +0.87]	+0.088	[-0.66, +0.84]	[-0.71, +0.89]	0.958 [0.920, 0.978]	0.962 [0.906, 0.985]
Rot (Full Finetuning)	+0.068	[-1.16, +1.30]	+0.068	[-1.04, +1.18]	[-1.17, +1.31]	0.889 [0.804, 0.938]	0.911 [0.811, 0.960]
Rot (Warmup-PerBlockLR)	+0.003	[-1.37, +1.37]	+0.003	[-1.21, +1.22]	[-1.38, +1.39]	0.857 [0.759, 0.917]	0.889 [0.791, 0.943]
Rot (Warmup-LRReset)	+0.026	[-1.36, +1.42]	+0.026	[-1.20, +1.25]	[-1.38, +1.43]	0.853 [0.749, 0.916]	0.887 [0.783, 0.943]
Rot (Fixed)	+0.007	[-1.45, +1.46]	+0.007	[-1.28, +1.29]	[-1.47, +1.48]	0.835 [0.723, 0.904]	0.874 [0.764, 0.934]

Tables S4A to S4D represent robustness analyses for slice level (n = 26) and patient level (n = 13): log-ratio Bland–Altman (multiplicative bias and LoA, applicable when measurements are positive and variance scales with magnitude), percentage Bland–Altman (relative bias and LoA), and Lin's CCC with both Fisher z-transformation and bootstrap (5000 resamples) 95% confidence intervals. Both Bland–Altman alternatives address the right-skewed distribution of viscoelastic values; bootstrap CIs provide a distribution-free complement to the Fisher CIs reported in Table V of the main paper.

TABLE S4A. SLICE-LEVEL (N = 26), STORAGE MODULUS (GD)

Model	Log-ratio bias	Log-ratio LoA (ratio)	%-bias	% LoA	CCC (Fisher CI)	CCC (Bootstrap CI)
Random init.	1.143	[0.58, 2.27]	+12.8%	[-53.2%, +78.8%]	0.896 [0.797, 0.949]	0.896 [0.390, 0.958]
AE (Full Finetuning)	1.092	[0.60, 2.00]	+8.3%	[-48.6%, +65.2%]	0.941 [0.874, 0.973]	0.941 [0.592, 0.983]
AE (Warmup-PerBlockLR)	1.104	[0.63, 1.93]	+9.5%	[-44.9%, +64.0%]	0.950 [0.894, 0.977]	0.950 [0.663, 0.981]
AE (Warmup-LRReset)	1.091	[0.65, 1.84]	+8.4%	[-42.8%, +59.5%]	0.953 [0.899, 0.979]	0.953 [0.698, 0.980]
AE (Fixed)	1.088	[0.65, 1.82]	+8.2%	[-41.8%, +58.3%]	0.958 [0.910, 0.981]	0.958 [0.732, 0.982]
Rot (Full Finetuning)	1.098	[0.69, 1.75]	+9.2%	[-36.6%, +55.0%]	0.961 [0.923, 0.980]	0.961 [0.794, 0.983]
Rot (Warmup-PerBlockLR)	1.068	[0.67, 1.71]	+6.5%	[-40.0%, +52.9%]	0.945 [0.894, 0.971]	0.945 [0.784, 0.977]
Rot (Warmup-LRReset)	1.086	[0.68, 1.74]	+8.1%	[-38.6%, +54.8%]	0.947 [0.896, 0.973]	0.947 [0.779, 0.977]
Rot (Fixed)	1.080	[0.67, 1.74]	+7.5%	[-39.6%, +54.6%]	0.936 [0.880, 0.967]	0.936 [0.777, 0.976]

TABLE S4B. SLICE-LEVEL (N = 26), LOSS MODULUS (GL)

Model	Log-ratio bias	Log-ratio LoA (ratio)	%-bias	% LoA	CCC (Fisher CI)	CCC (Bootstrap CI)
Random init.	1.243	[0.56, 2.77]	+20.7%	[-55.2%, +96.6%]	0.835 [0.713, 0.908]	0.835 [0.294, 0.956]
AE (Full Finetuning)	1.176	[0.57, 2.42]	+15.2%	[-51.5%, +82.0%]	0.942 [0.884, 0.971]	0.942 [0.419, 0.979]
AE (Warmup-PerBlockLR)	1.183	[0.61, 2.28]	+16.1%	[-46.9%, +79.1%]	0.945 [0.901, 0.970]	0.945 [0.542, 0.970]
AE (Warmup-LRReset)	1.186	[0.64, 2.20]	+16.4%	[-43.3%, +76.1%]	0.958 [0.920, 0.978]	0.958 [0.580, 0.979]
AE (Fixed)	1.165	[0.63, 2.14]	+14.8%	[-44.2%, +73.8%]	0.958 [0.920, 0.978]	0.958 [0.615, 0.978]
Rot (Full Finetuning)	1.181	[0.64, 2.18]	+16.3%	[-43.4%, +75.9%]	0.889 [0.804, 0.938]	0.889 [0.640, 0.969]
Rot (Warmup-PerBlockLR)	1.133	[0.63, 2.03]	+12.3%	[-44.8%, +69.4%]	0.857 [0.759, 0.917]	0.857 [0.697, 0.968]
Rot (Warmup-LRReset)	1.159	[0.63, 2.12]	+14.5%	[-44.7%, +73.6%]	0.853 [0.749, 0.916]	0.853 [0.669, 0.968]
Rot (Fixed)	1.146	[0.63, 2.09]	+13.4%	[-45.1%, +71.9%]	0.835 [0.723, 0.904]	0.835 [0.672, 0.966]

TABLE S4C. PATIENT-LEVEL (N = 13), STORAGE MODULUS (Gd)

Model	Log-ratio bias	Log-ratio LoA (ratio)	%-bias	% LoA	CCC (Fisher CI)	CCC (Bootstrap CI)
Random init.	1.149	[0.60, 2.20]	+13.4%	[-49.3%, +76.1%]	0.917 [0.793, 0.968]	0.917 [-0.229, 0.974]
AE (Full Finetuning)	1.098	[0.61, 1.99]	+8.9%	[-48.1%, +65.8%]	0.951 [0.849, 0.985]	0.951 [-0.277, 0.988]
AE (Warmup-PerBlockLR)	1.104	[0.63, 1.94]	+9.5%	[-45.2%, +64.3%]	0.952 [0.855, 0.985]	0.952 [-0.194, 0.989]
AE (Warmup-LRReset)	1.091	[0.65, 1.84]	+8.5%	[-42.4%, +59.4%]	0.956 [0.863, 0.986]	0.956 [-0.099, 0.986]
AE (Fixed)	1.089	[0.65, 1.81]	+8.3%	[-41.8%, +58.3%]	0.960 [0.877, 0.988]	0.960 [-0.113, 0.987]
Rot (Full Finetuning)	1.099	[0.70, 1.72]	+9.3%	[-34.6%, +53.2%]	0.966 [0.913, 0.987]	0.966 [0.165, 0.989]
Rot (Warmup-PerBlockLR)	1.071	[0.69, 1.66]	+6.8%	[-36.4%, +49.9%]	0.957 [0.901, 0.982]	0.957 [0.260, 0.981]
Rot (Warmup-LRReset)	1.089	[0.70, 1.70]	+8.4%	[-35.9%, +52.7%]	0.960 [0.904, 0.984]	0.960 [0.187, 0.984]
Rot (Fixed)	1.083	[0.70, 1.68]	+7.9%	[-35.8%, +51.5%]	0.953 [0.896, 0.979]	0.953 [0.248, 0.975]

TABLE S4D. PATIENT-LEVEL (N = 13), LOSS MODULUS (GL)

Model	Log-ratio bias	Log-ratio LoA (ratio)	%-bias	% LoA	CCC (Fisher CI)	CCC (Bootstrap CI)
Random init.	1.251	[0.59, 2.67]	+21.4%	[-50.8%, +93.6%]	0.877 [0.760, 0.939]	0.877 [-0.261, 0.912]
AE (Full Finetuning)	1.184	[0.58, 2.40]	+16.0%	[-51.2%, +83.2%]	0.953 [0.874, 0.983]	0.953 [-0.367, 0.980]
AE (Warmup-PerBlockLR)	1.182	[0.61, 2.28]	+16.1%	[-47.1%, +79.3%]	0.949 [0.882, 0.979]	0.949 [-0.301, 0.968]
AE (Warmup-LRReset)	1.186	[0.64, 2.19]	+16.5%	[-42.6%, +75.7%]	0.962 [0.906, 0.985]	0.962 [-0.216, 0.980]
AE (Fixed)	1.165	[0.64, 2.13]	+14.8%	[-43.9%, +73.6%]	0.962 [0.906, 0.985]	0.962 [-0.270, 0.977]
Rot (Full Finetuning)	1.182	[0.65, 2.14]	+16.3%	[-41.6%, +74.2%]	0.911 [0.811, 0.960]	0.911 [-0.023, 0.927]
Rot (Warmup-PerBlockLR)	1.137	[0.66, 1.95]	+12.6%	[-40.7%, +65.9%]	0.889 [0.791, 0.943]	0.889 [0.141, 0.924]
Rot (Warmup-LRReset)	1.162	[0.65, 2.06]	+14.7%	[-41.6%, +71.0%]	0.887 [0.783, 0.943]	0.887 [0.059, 0.909]
Rot (Fixed)	1.150	[0.66, 2.00]	+13.7%	[-40.8%, +68.3%]	0.874 [0.764, 0.934]	0.874 [0.127, 0.917]

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